

S2 Major Module Coursework

Written Answers

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Question 1 – Accident data

The mean rate of accidents over the full observation period is

$$\bar{x} = \frac{N}{L} = \frac{191}{40550} \approx 4.710 \times 10^{-3} \text{ accidents per day.}$$

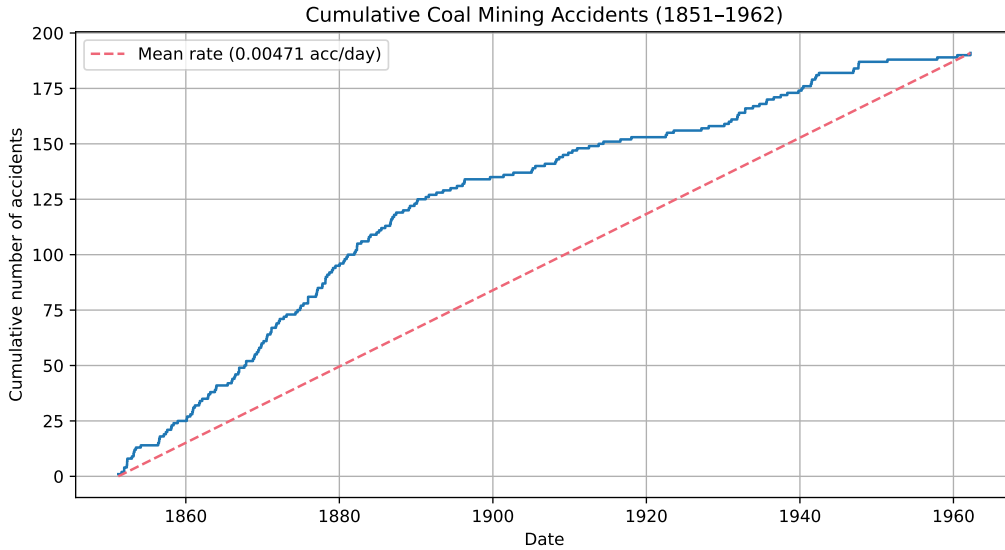


Figure 1: Cumulative number of mining accidents (1851–1962). The dashed line shows the overall mean rate.

Question 2 – Priors

(a) Comparing change-point priors

The plain order statistics prior draws k uniform points on $[0, L]$ and sorts them. The even-numbered order statistics prior instead draws $2k + 1$ uniform points and retains every second one (the even-numbered order statistics). Because each retained change point has an “escort” point on either side, the even-numbered order statistics prior discourages change points from clustering together or from sitting near the boundaries. In effect, the escort points act as spacers: a gap g_j between adjacent change points enters the density as a factor of g_j , penalising small gaps.

Figure 2 illustrates this for $k = 4$. The marginal distributions of individual change points are more peaked and pushed away from the endpoints under the even-numbered prior. The plain

order statistics prior, by contrast, allows change points to cluster more freely and to approach 0 or L . The even-numbered prior therefore favours configurations where change points are more evenly spread across the observation window, which is a sensible default when there is no prior reason to expect clustering.

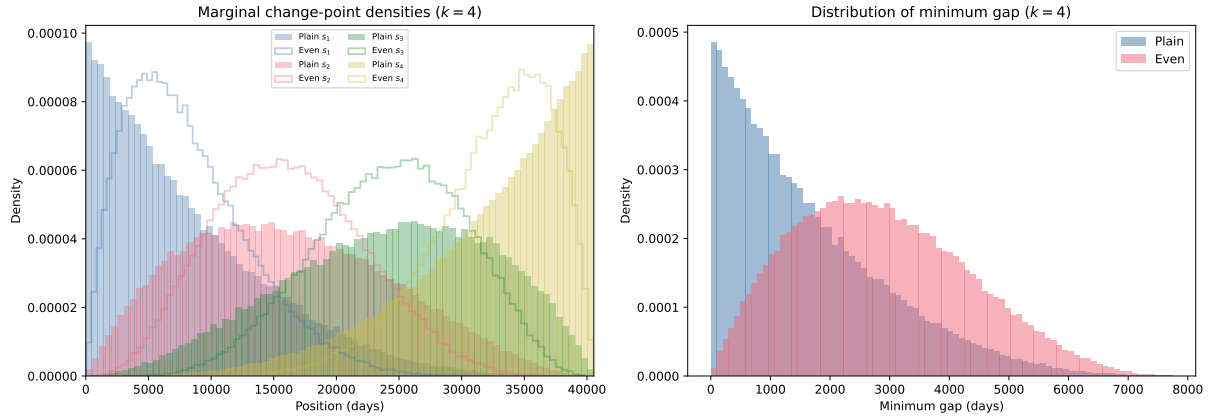


Figure 2: Marginal distributions of individual change points for $k = 4$ under the plain order statistics prior (left) and even-numbered order statistics prior (right).

(b) Derivation of the even-numbered order statistics prior

The PDF of the plain order statistics of $2k + 1$ uniform random variables on $[0, 1]$ is

$$P(u_{(1)}, u_{(2)}, \dots, u_{(2k+1)}) = (2k + 1)!$$

on the region $0 < u_{(1)} < u_{(2)} < \dots < u_{(2k+1)} < 1$.

The even-numbered order statistics are $\hat{s}_j = u_{(2j)}$ for $j = 1, \dots, k$. To obtain their marginal density, we integrate out the $k + 1$ odd-indexed variables $u_{(1)}, u_{(3)}, \dots, u_{(2k+1)}$.

Each odd variable $u_{(2j-1)}$ is constrained to the interval $[\hat{s}_{j-1}, \hat{s}_j]$ (using the conventions $\hat{s}_0 = 0$, $\hat{s}_{k+1} = 1$). Since the integrand is constant in each odd variable, each integral contributes a factor equal to the length of its support:

$$\int_{\hat{s}_{j-1}}^{\hat{s}_j} du_{(2j-1)} = \hat{s}_j - \hat{s}_{j-1} = \hat{g}_j.$$

Performing all $k + 1$ such integrals (for $u_{(1)}, u_{(3)}, \dots, u_{(2k+1)}$) gives

$$P(\hat{s}_1, \hat{s}_2, \dots, \hat{s}_k) = (2k + 1)! \prod_{j=1}^{k+1} \hat{g}_j \propto \prod_{j=1}^{k+1} \hat{g}_j,$$

where $\hat{g}_j = \hat{s}_j - \hat{s}_{j-1}$ are the gaps on the unit interval.

Rescaling to the observation interval via $s_j = L\hat{s}_j$ gives $g_j = L\hat{g}_j$, so

$$\pi(s_1, s_2, \dots, s_k \mid M_k) \propto \prod_{j=1}^{k+1} g_j.$$

Question 3 – The constant rate model, M_0

(a) Prior and posterior for h_0 ; evidence Z_0

Under M_0 the rate is constant at h_0 with prior $h_0 \sim \Gamma(1, 200)$, i.e. $\pi(h_0) = 200 e^{-200h_0}$. The likelihood is $\mathcal{L} = h_0^N e^{-h_0 L}$, so the unnormalised posterior is $h_0^N e^{-h_0(L+\beta)}$, which is a Gamma distribution with shape $\alpha + N = 192$ and rate $\beta + L = 40750$.

Figure 3 compares the prior and posterior. The prior is an exponential distribution broadly spread over small rates, while the posterior is sharply concentrated around $h_0 \approx N/(L + \beta) \approx 4.71 \times 10^{-3}$ accidents per day, essentially the observed mean rate. The data are highly informative: the posterior is orders of magnitude narrower than the prior.

The evidence is computed analytically:

$$Z_0 = \int_0^\infty \pi(h_0) \mathcal{L}(\{I_i\} | h_0) dh_0 = \frac{\beta^\alpha \Gamma(\alpha + N)}{(\beta + L)^{\alpha+N}} \implies \log Z_0 = -1217.09.$$

This was verified by numerical integration, yielding identical results.

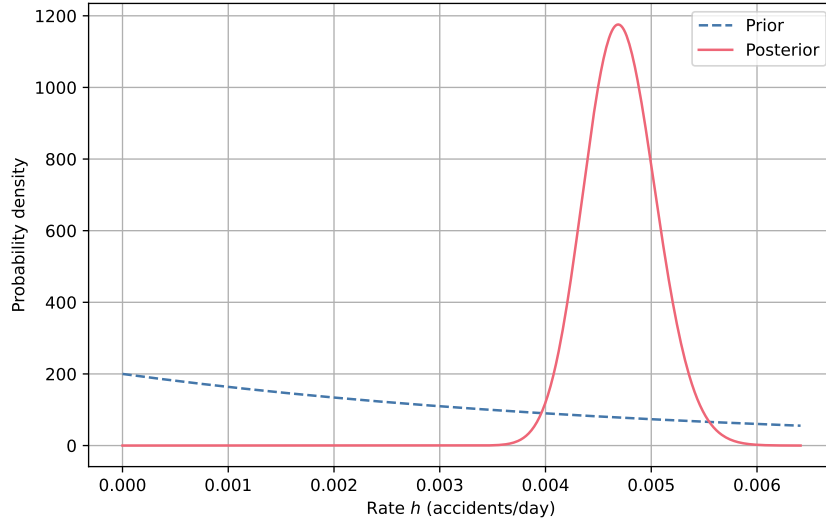


Figure 3: Prior and normalised posterior for the constant accident rate h_0 under model M_0 .

Question 4 – The 1 change-point model, M_1

(a) MCMC posterior sampling

We sampled the M_1 posterior using **emcee** (Foreman-Mackey et al., 2013), an affine-invariant ensemble sampler. Unlike gradient-based methods (e.g. NUTS), **emcee** requires only point-wise evaluation of the log-posterior, which suits the piecewise-constant likelihood that is non-differentiable at s_1 . We used 32 walkers initialised in a tight ball around a reasonable starting point ($h_0 \approx 0.009$, $h_1 \approx 0.003$, $s_1 \approx 14000$ days). After a burn-in of 2000 steps the sampler was reset and run for a further 10000 production steps.

The priors are $h_0, h_1 \sim \Gamma(1, 200)$ and $s_1/L \sim \text{Beta}(2, 2)$ (the even-numbered order statistics prior for $k = 1$).

Convergence was assessed via: (i) visual inspection of trace plots (Figure 4, showing good mixing with no trends); (ii) integrated autocorrelation times ($\tau_{h_0} \approx 49$, $\tau_{h_1} \approx 49$, $\tau_{s_1} \approx 48$

steps), yielding roughly 6500 effective samples; and (iii) the mean acceptance fraction of 0.52, within the recommended range for `emcee`.

The posterior medians and 68% credible intervals are $h_0 = 0.0086 \pm 0.0008$, $h_1 = 0.0025 \pm 0.0003$, and $s_1 = 14\,362 \pm 770$ days (corresponding to ~ 1890). The corner plot (Figure 5) shows a clear separation between h_0 and h_1 , confirming that the data strongly favour a decrease in accident rate.

We note that the $\Gamma(1, 200)$ priors on h_0 and h_1 are conjugate to the Poisson process likelihood. For fixed s_1 the likelihood factorises, so each height can be analytically marginalised: $\int_0^\infty h_0^{n_0} e^{-h_0(s_1+\beta)} \beta^\alpha dh_0 = \beta^\alpha \Gamma(\alpha + n_0) / (\beta + s_1)^{\alpha+n_0}$, and similarly for h_1 . This yields a one-dimensional marginal posterior on s_1 alone, from which h_0 and h_1 samples can be drawn conditionally as $h_0 | s_1 \sim \Gamma(\alpha + n_0, \beta + s_1)$. Such a collapsed (Rao–Blackwellised) approach would reduce the sampling problem from three dimensions to one, but we retain the full `emcee` sampler here for generality.

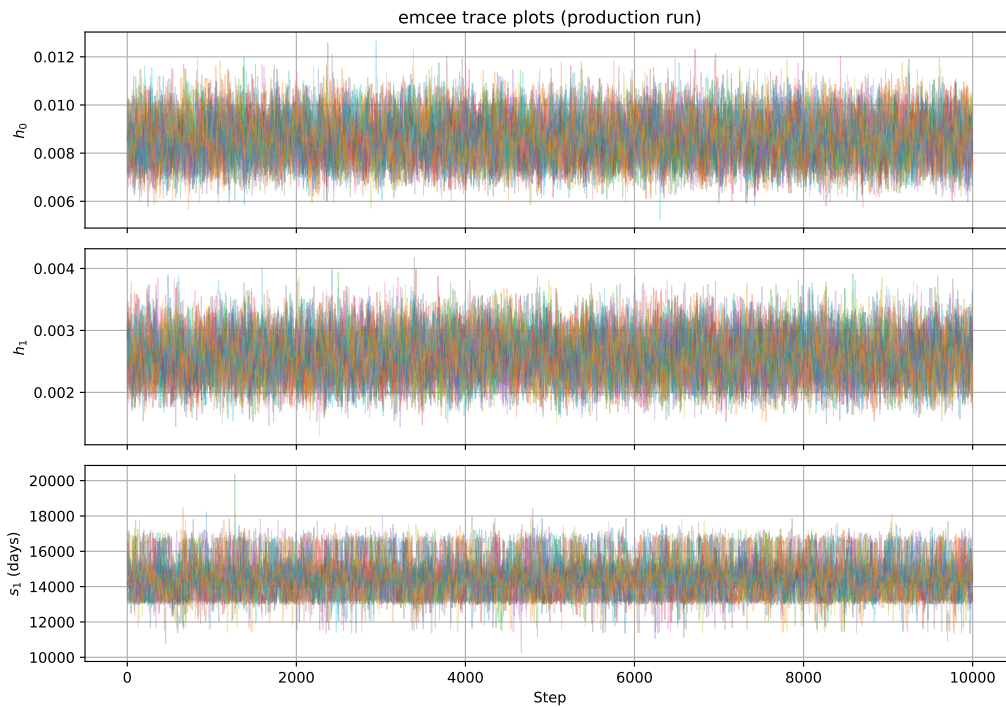


Figure 4: Trace plots for the M_1 parameters from the `emcee` production run.

(b) Savage–Dickey density ratio

The Savage–Dickey density ratio provides the Bayes factor between nested models without computing evidences directly. Model M_0 is recovered from M_1 by setting $h_0 = h_1$ (equivalently $\delta = h_0 - h_1 = 0$). The Bayes factor is then

$$B_{01} = \frac{Z_0}{Z_1} = \frac{p(\delta = 0 | \{I_i\}, M_1)}{\pi(\delta = 0 | M_1)},$$

i.e. the ratio of the marginal posterior to prior density of δ evaluated at $\delta = 0$, both computed under M_1 . This avoids any explicit evidence calculation.

The Savage–Dickey ratio requires evaluating the numerator $p(\delta = 0 | \{I_i\}, M_1)$. In practice this density must be estimated from MCMC samples drawn from the M_1 posterior (e.g. via a histogram or kernel density estimate at $\delta = 0$). However, the M_1 posterior strongly favours

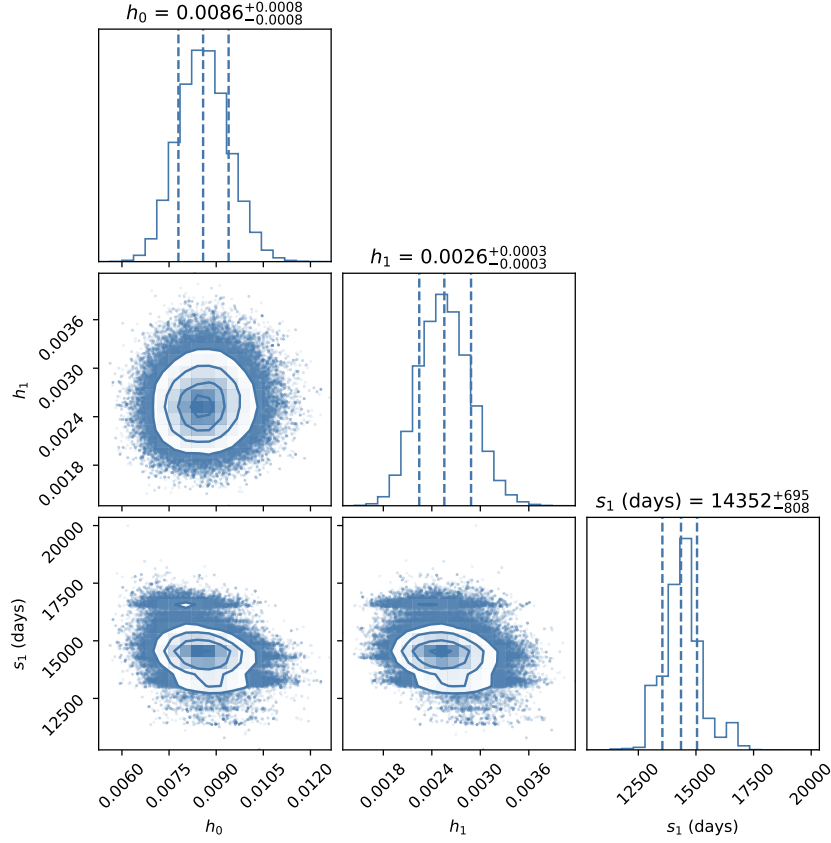


Figure 5: Corner plot for the M_1 posterior.

distinct rates before and after the change point: the marginal posterior of δ concentrates around $\delta \approx 0.006$, far from zero (Figure 6). Virtually no MCMC samples fall near $\delta = 0$, so any density estimate there is dominated by noise.

If evaluated, the ratio would give $B_{01} \approx 0$ (i.e. $B_{10} \rightarrow \infty$), qualitatively confirming M_1 is favoured. However, the KDE value at $\delta = 0$ depends entirely on bandwidth and random fluctuations and could vary by many orders of magnitude, making the Bayes factor estimate quantitatively useless.

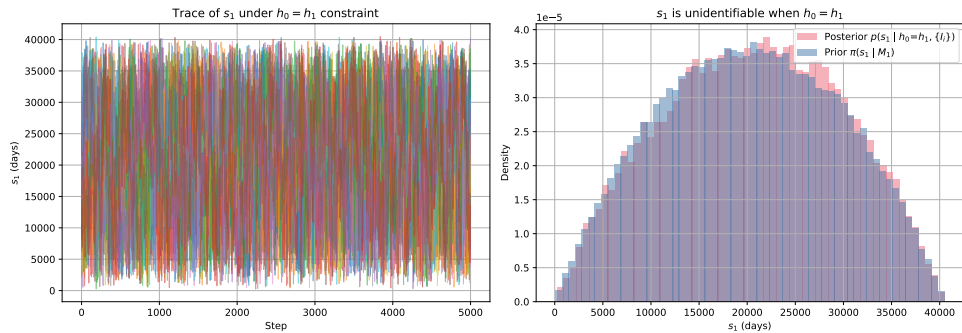


Figure 6: Marginal posterior of $\delta = h_0 - h_1$ under M_1 . The nested point $\delta = 0$ lies deep in the tail where no MCMC samples reach, making the Savage–Dickey numerator impossible to estimate reliably.

(c) Laplace approximation

The Laplace approximation estimates the evidence by fitting a multivariate Gaussian to the log-posterior at its mode:

$$Z_1 \approx (2\pi)^{d/2} |\hat{\Sigma}|^{1/2} \mathcal{L}(\hat{\theta}) \pi(\hat{\theta}),$$

where $d = 3$, $\hat{\theta}$ is the MAP estimate, and $\hat{\Sigma}$ is the inverse of the negative Hessian of the log-posterior at $\hat{\theta}$.

The difficulty is that the M_1 log-likelihood contains a hard step function at s_1 : an event at time y_i contributes $\log h_0$ if $y_i < s_1$ and $\log h_1$ otherwise. The log-likelihood is therefore piecewise linear in s_1 between adjacent event times, with discontinuous jumps whenever s_1 crosses an event time. The Hessian $\partial^2 \log P / \partial s_1^2$ is zero almost everywhere with delta-function singularities at each event time; it is not well defined. The Gaussian approximation is therefore invalid: the posterior is not smooth in s_1 , and the Laplace approximation breaks down.

(d) Nested sampling for M_1 evidence

We used `dynesty`'s static nested sampler with 1000 live points and a stopping criterion of $\Delta \log Z < 0.01$. The prior transform maps from the unit cube to the physical parameter space via the inverse CDFs of the $\Gamma(1, 200)$ priors for h_0, h_1 and the Beta(2, 2) prior (scaled to $[0, L]$) for s_1 . The log-likelihood is the same piecewise-constant Poisson process expression used in the MCMC; `dynesty` requires no gradients, so the discontinuity in s_1 poses no difficulty. Multi-ellipsoidal decomposition was used for bounding the live-point distribution.

The results are:

$$\begin{aligned} \log Z_1 &= -1187.54 \pm 0.12, \\ \log Z_0 &= -1217.09 \quad (\text{analytic, from Q3a}), \\ \log B_{10} &= \log(Z_1/Z_0) = 29.56, \quad B_{10} \approx 6.85 \times 10^{12}. \end{aligned}$$

The prior odds are $\pi(M_1)/\pi(M_0) = \lambda = 3$ (from the truncated Poisson prior on k), giving

$$\text{Posterior odds} = B_{10} \times \frac{\pi(M_1)}{\pi(M_0)} \approx 2.05 \times 10^{13}.$$

The data overwhelmingly favour M_1 over M_0 .

Question 5 – The k change-point models, M_k

(b) RJMCMC trace and posterior on k

The reversible jump MCMC sampler was run for 10^5 iterations. Figure 7 shows the trace plot for the model index k and the posterior distribution on k . The trace plot confirms that the chain explores the model space thoroughly, visiting a wide range of k values with rapid mixing.

The maximum *a posteriori* (MAP) number of change points is $\mathbf{k} = \mathbf{3}$, with posterior probability $P(k = 3 \mid \{I_i\}) = 0.279$. The top five values are $k = 3$ (0.279), $k = 2$ (0.261), $k = 4$ (0.218), $k = 5$ (0.104), and $k = 1$ (0.087).

(c) Inferred rate with uncertainty

Figure 8 shows the model-averaged inferred accident rate $x(t)$ with 50% and 90% credible regions.

Calculating the uncertainty regions. For each post-burn-in RJMCMC sample, the state encodes a complete step-function rate $x(t)$ with a particular number of change points k , change-point positions, and segment rates. Each sample's rate function was evaluated on a dense grid

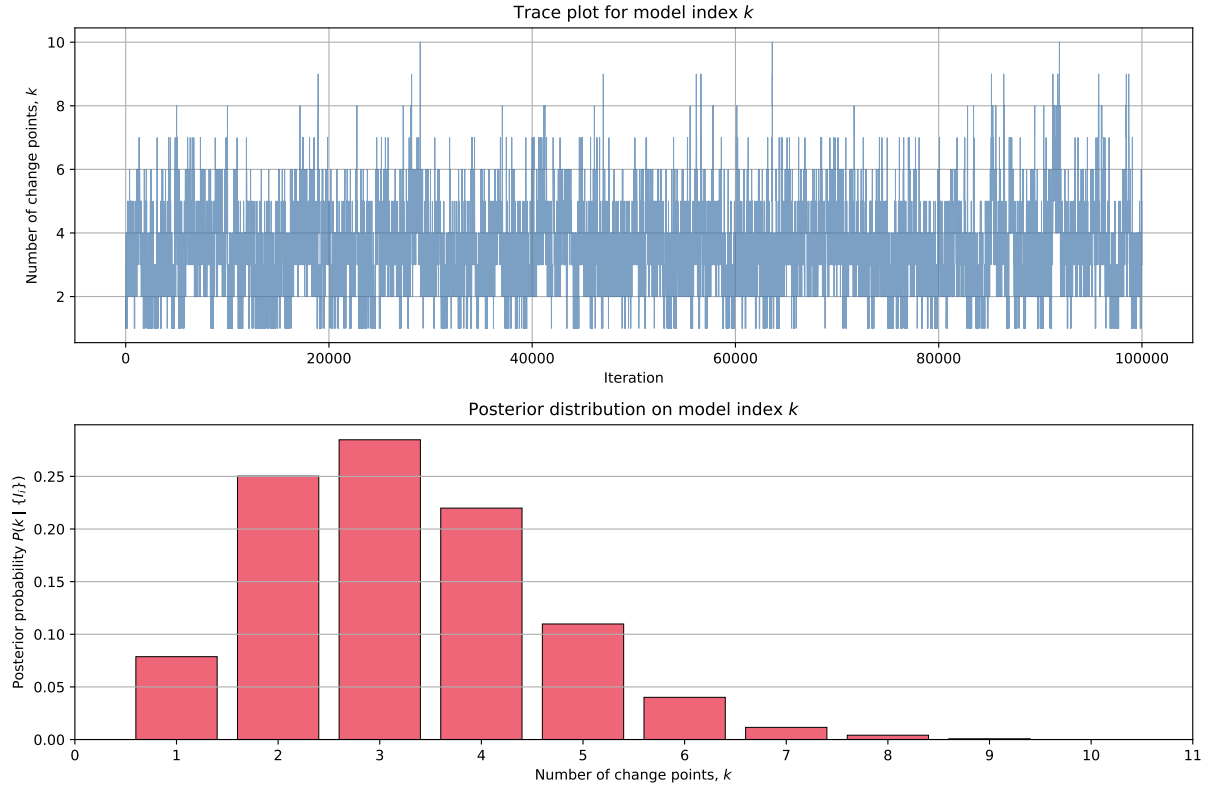


Figure 7: Top: trace plot for the model index k . Bottom: posterior distribution on k .

of 1000 time points spanning the observation period. This produces, at each grid point, an ensemble of rate values drawn from the full model-averaged posterior; samples with different k contribute naturally since the RJMCMC chain explores the joint space of all models. The 50% credible region is the interval between the 25th and 75th percentiles of this ensemble at each time point, and the 90% region spans the 5th to 95th percentiles. The median is the pointwise 50th percentile. These are pointwise credible intervals, not simultaneous bands.

Declaration of use of autogeneration tools

Claude Code (Anthropic) was used as a coding assistant throughout this coursework for: code scaffolding, debugging, plotting, and drafting written explanations. All outputs were reviewed, verified, and edited by the author. The mathematical derivations and scientific interpretations reflect the author's own understanding.

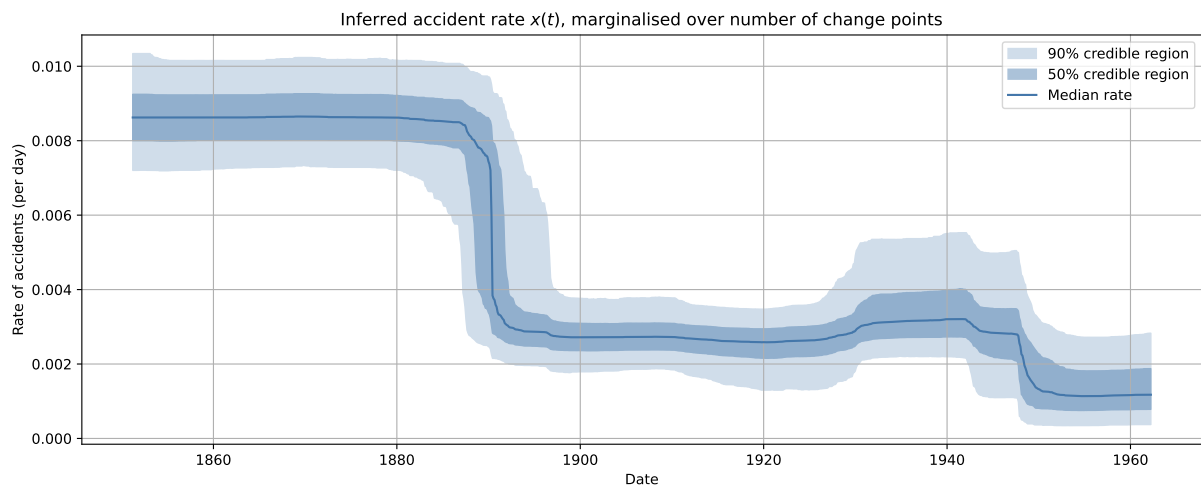


Figure 8: Inferred accident rate $x(t)$ marginalised over the number of change points, with 50% and 90% pointwise credible regions.